Pimpri Chinchwad Education Trust's
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Assignment 3: Statistical Hypothesis Testing (Data Analysis Exercise)

Research Methodology | **Group No:** 19

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Major Project: BhashaBridge: Camera-Only, Continuous Real-Time Marathi Sign Language Recognition using MediaPipe landmarks and a hybrid 1D-CNN + BiLSTM + temporal-attention model

Tests applied: Independent-Samples (Welch) t-Test, One-Way ANOVA, Chi-Square Test of Independence

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0. Purpose and Data Source

This assignment applies the three tests from the course guide (Independent-Samples t-Test, One-Way ANOVA and Chi-Square Test of Independence) to data related to our major project, BhashaBridge. The assignment allows a sample dataset to be assumed when real project data is not available.

Important note on the data. BhashaBridge is still at the design and validation-planning stage, so no measured recognition results exist yet. The three datasets below are therefore **assumed, illustrative samples**. They were designed to match the validation plan in our Systematic Literature Review (Idle class with confidence debounce, lighting and domain shift, and a small signing vocabulary). They are **not measured results** and must not be quoted as the performance of the system. Every statistic in this document is calculated correctly from the assumed numbers, so the method can be reused without change when real data is collected.

Table 1. Assumed datasets used in this assignment.

Dataset	Used for	Variables	Groups and sample size
A	Welch t-test	False sign outputs per minute during idle (resting-hand) recordings (numerical)	Baseline classifier without Idle class vs proposed classifier with Idle class + 85% debounce; 10 recordings each (n = 20)
B	One-way ANOVA	Recognition accuracy (%) of a test session (numerical)	Three lighting conditions: Bright, Normal, Dim; 8 sessions each (N = 24)
C	Chi-square	Sign category and recognition outcome (both categorical)	Greetings, Medical, Numbers vs Correct / Incorrect; 100 attempts per category (N = 300)

General workflow followed for every test: define the question and variable types, state H0 and H1, set $\alpha = 0.05$, check assumptions, run the test, report the statistic, degrees of freedom and p-value, and interpret the result in the context of the project.

1. Independent-Samples t-Test (Welch)

1.1 Purpose and research question

Our SLR (research gap G3) argues that a classifier trained only on sign classes must label every window as a sign, even when the signer is resting. The proposed design adds an explicit Idle class and an 85% confidence debounce filter. **Does the proposed design produce a different mean number of false sign outputs per minute during idle periods than the baseline classifier?**

Item	Detail
Outcome variable	False sign outputs per minute during an idle recording (numerical)
Grouping variable	Classifier design (two independent groups)
Groups	Baseline (no Idle class), n = 10; Proposed (Idle class + debounce), n = 10
H0	μ(Baseline) = μ(Proposed)
H1	μ(Baseline) ≠ μ(Proposed) (two-tailed)
Significance level	$\alpha = 0.05$

1.2 Sample data (assumed, Dataset A)

Table 2. False sign outputs per minute, one value per idle recording.

Recording	Baseline (no Idle class)	Proposed (Idle + debounce)
1	6.2	3.1
2	8.9	4.8
3	5.1	2.0
4	9.4	4.4
5	6.9	3.6
6	7.8	2.5
7	4.6	5.0
8	8.4	3.0
9	5.9	4.1
10	7.0	2.2
Mean	7.02	3.47
SD (n − 1)	1.61	1.08

1.3 Manual calculation

1. Means: 7.02 and 3.47, so the observed difference is 7.02 − 3.47 = 3.55.
2. Sample variances: s^2(Baseline) = 2.577 and s^2(Proposed) = 1.162.
3. Standard error = $\sqrt{(2.577/10 + 1.162/10)} = \sqrt{0.3739} = 0.6115$.
4. t = 3.55 / 0.6115 = **5.81**.
5. Welch degrees of freedom = $(0.2577 + 0.1162)^2 / [0.2577^2/9 + 0.1162^2/9]$ = **15.75**.
6. The critical value is t(0.975, 15.75) ≈ 2.12. Since |5.81| > 2.12, H0 is rejected. The p-value is 0.000029.

1.4 Conduct the test in Excel

1. Enter Baseline values in B5:B14 and Proposed values in C5:C14 (see the attached workbook, sheet “t-test”).
2. Enable Analysis ToolPak: File → Options → Add-ins → Excel Add-ins → Go → tick Analysis ToolPak.
3. Data → Data Analysis → t-Test: Two-Sample Assuming Unequal Variances.
4. Select both ranges, Hypothesized Mean Difference = 0, Alpha = 0.05, choose an output range.
5. Read t Stat (5.81), df (15.75) and P(T<=t) two-tail (0.000029). Formula check: =T.TEST(B5:B14,C5:C14,2,3).

Descriptive formulas: =AVERAGE(B5:B14), =AVERAGE(C5:C14), =STDEV.S(B5:B14), =STDEV.S(C5:C14).

1.5 Python code

```
import numpy as np
from scipy import stats

baseline = np.array([6.2, 8.9, 5.1, 9.4, 6.9, 7.8, 4.6, 8.4, 5.9, 7.0])
proposed = np.array([3.1, 4.8, 2.0, 4.4, 3.6, 2.5, 5.0, 3.0, 4.1, 2.2])
print("Baseline mean:", baseline.mean(), "SD:", baseline.std(ddof=1))
print("Proposed mean:", proposed.mean(), "SD:", proposed.std(ddof=1))

# assumption checks
print(stats.shapiro(baseline), stats.shapiro(proposed))
print(stats.levene(baseline, proposed))

t_stat, p_value = stats.ttest_ind(baseline, proposed, equal_var=False)  # Welch
print("t-statistic:", t_stat, "p-value:", p_value)

if p_value < 0.05:
    print("Reject H0: evidence of a difference in mean false outputs.")
else:
    print("Fail to reject H0: insufficient evidence of a difference.")
```

1.6 Result and interpretation (on the assumed data)

Baseline mean = 7.02 (SD 1.61); Proposed mean = 3.47 (SD 1.08). Welch’s test gives **t(15.75) = 5.81, p < .001** (p = 0.000029, two-sided). The mean difference is 3.55 false outputs per minute, 95% CI [2.25, 4.85]. Cohen’s d = 2.60, a very large effect.

Since p < 0.05, **reject H0**. On this assumed sample the proposed design produces significantly fewer false sign outputs during idle periods, about 51% fewer than the baseline (3.55 / 7.02). If real recordings looked like this, it would support the SLR argument that an explicit Idle class and debounce filter reduce temporal hallucination.

Suggested reporting sentence: “A Welch t-test showed that the proposed classifier produced significantly fewer false outputs per minute (M = 3.47, SD = 1.08) than the baseline (M = 7.02, SD = 1.61), t(15.75) = 5.81, p < .001, d = 2.60.”

1.7 Assumptions and cautions

- **Independence:** each value comes from a separate idle recording, so the two groups are independent (assumed). If both classifiers were run on the same recordings, a paired t-test would be the correct choice.
- **Normality:** Shapiro–Wilk gives W = 0.969, p = 0.879 (Baseline) and W = 0.942, p = 0.573 (Proposed), so normality is not rejected.
- **Equal variances:** Levene’s test gives p = 0.262. Welch’s test does not need equal variances anyway, and the Student t-test gives the same decision, t(18) = 5.81, p < .001.

- With n = 10 per group the test has limited precision, and the numbers are assumed, not measured.

2. One-Way ANOVA

2.1 Purpose and research question

The SLR identifies spatial bias and domain shift (research gap G2). Lighting is one of the most common causes of landmark-detection errors. **Does mean recognition accuracy differ across Bright, Normal and Dim lighting conditions?**

Item	Detail
Outcome variable	Session recognition accuracy in % (numerical)
Factor / grouping variable	Lighting condition
Groups	Bright, Normal, Dim (8 sessions each, N = 24)
H0	μ(Bright) = μ(Normal) = μ(Dim)
H1	At least one population mean differs
Significance level	α = 0.05

2.2 Sample data (assumed, Dataset B)

Table 3. Recognition accuracy (%) per test session.

Session	Bright	Normal	Dim
1	94.2	90.5	85.3
2	91.8	88.2	88.9
3	95.6	92.8	83.6
4	92.5	87.7	89.4
5	90.9	90.1	86.2
6	94.7	92.0	82.9
7	92.0	88.6	88.1
8	93.4	90.9	85.8
Mean	93.14	90.10	86.28
SD	1.61	1.83	2.38

2.3 Formula concept and manual calculation

F = MS(between) / MS(within), where MS(between) = SS(between) / (k − 1) and MS(within) = SS(within) / (N − k). Here k = 3 and N = 24, so df = (2, 21). The grand mean is 89.84.

Table 4. ANOVA summary table.

Source	SS	df	MS	F	p-value
Between groups	189.20	2	94.60	24.46	0.0000033
Within groups	81.23	21	3.87		
Total	270.44	23			

SS(between) = $8(93.14 - 89.84)^2 + 8(90.10 - 89.84)^2 + 8(86.28 - 89.84)^2 = 189.20$. SS(within) is the sum of squared deviations inside each group = 81.23. Effect size $\eta^2 = 189.20 / 270.44 = 0.70$.

2.4 Conduct ANOVA in Excel

1. Enter the three columns with headers Bright, Normal, Dim in B4:D12 (workbook sheet “ANOVA”).

2. Data → Data Analysis → ANOVA: Single Factor.
3. Select the input range, Grouped By: Columns, tick Labels in First Row, Alpha = 0.05.
4. Read F (24.46), P-value (0.0000033) and the between / within degrees of freedom (2 and 21).
5. Because p < 0.05, run a post-hoc test (Tukey HSD) to find which pairs differ (see 2.5).

2.5 Python code

```
import numpy as np
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd

bright = np.array([94.2, 91.8, 95.6, 92.5, 90.9, 94.7, 92.0, 93.4])
normal = np.array([90.5, 88.2, 92.8, 87.7, 90.1, 92.0, 88.6, 90.9])
dim    = np.array([85.3, 88.9, 83.6, 89.4, 86.2, 82.9, 88.1, 85.8])

print(stats.levene(bright, normal, dim))          # equal variances?
f_stat, p_value = stats.f_oneway(bright, normal, dim)
print("F:", f_stat, "p:", p_value)

print(stats.kruskal(bright, normal, dim))         # non-parametric check

# post-hoc test, needed because ANOVA is significant
scores = np.concatenate([bright, normal, dim])
groups = ["Bright"]*8 + ["Normal"]*8 + ["Dim"]*8
print(pairwise_tukeyhsd(scores, groups, alpha=0.05))
```

2.6 Result and interpretation (on the assumed data)

Group means: Bright = 93.14%, Normal = 90.10%, Dim = 86.28%. One-way ANOVA gives **$F(2, 21) = 24.46$, $p < .001$, $\eta^2 = 0.70$**. Since p < 0.05, **reject H0**: at least one lighting condition has a different mean accuracy.

Table 5. Tukey HSD post-hoc comparisons (α = 0.05).

Comparison	Mean difference	95% CI	Adjusted p	Decision
Bright − Normal	3.04	[0.56, 5.52]	0.015	Significant
Bright − Dim	6.86	[4.38, 9.34]	< .001	Significant
Normal − Dim	3.83	[1.35, 6.30]	0.002	Significant

Every pair differs, and accuracy falls steadily as lighting gets worse. In project terms, if real sessions behaved like this the SLR domain-shift concern would be confirmed, and lighting augmentation would be needed in training. The practical drop between Bright and Dim lighting is almost 7 percentage points.

Suggested reporting sentence: "A one-way ANOVA showed a significant effect of lighting on recognition accuracy, $F(2, 21) = 24.46$, $p < .001$, $\eta^2 = .70$. Tukey HSD tests showed that all three conditions differed from each other (all $p < .05$), with accuracy highest in Bright and lowest in Dim lighting."

2.7 Assumptions and cautions

- **Independence:** each value is a separate session (assumed). If the same signers appear in all three conditions, a repeated-measures ANOVA is more appropriate.
- **Normality:** Shapiro–Wilk p = 0.882 (Bright), 0.737 (Normal), 0.648 (Dim), so normality is not rejected.
- **Homogeneity of variance:** Levene's test p = 0.567, so equal variances are acceptable.
- **Robustness check:** the Kruskal–Wallis test agrees, $H(2) = 16.26$, $p < .001$.

3. Chi-Square Test of Independence

3.1 Purpose and research question

The planned MSL vocabulary will contain different kinds of signs. **Is the recognition outcome (correct or incorrect) associated with the category of the sign?**

Item	Detail
Variable 1	Sign category: Greetings / Medical / Numbers (assumed vocabulary groups)
Variable 2	Recognition outcome: Correct / Incorrect
H0	Sign category and recognition outcome are independent
H1	Sign category and recognition outcome are associated
Significance level	$\alpha = 0.05$

3.2 Observed frequency table (assumed, Dataset C)

Table 6. Observed counts of recognition attempts.

Sign category	Correct	Incorrect	Total	Error rate
Greetings	93	7	100	7%
Medical	85	15	100	15%
Numbers	78	22	100	22%
Total	**256**	**44**	**300**	**14.7%**

3.3 Expected frequencies and chi-square formula

Expected = (row total × column total) / grand total. Each category has 100 attempts, so for every category: Correct E = (100 × 256) / 300 = 85.33 and Incorrect E = (100 × 44) / 300 = 14.67. Chi-square = $\Sigma\ (O - E)^2 / E$.

Table 7. Observed, expected and chi-square contributions.

Cell	O	E	$(O-E)^2/E$	Std. residual
Greetings, Correct	93	85.33	0.689	+0.83
Greetings, Incorrect	7	14.67	4.008	−2.00
Medical, Correct	85	85.33	0.001	−0.04
Medical, Incorrect	15	14.67	0.008	+0.09
Numbers, Correct	78	85.33	0.630	−0.79
Numbers, Incorrect	22	14.67	3.667	+1.91
Total χ^2			**9.002**	

Degrees of freedom = (3 − 1) × (2 − 1) = 2. The critical value at $\alpha = 0.05$ is 5.99, and 9.00 > 5.99, so H0 is rejected.

3.4 Conduct the test in Excel

1. Enter the 3 × 2 observed table in B5:C7 (workbook sheet “Chi-square”), without totals in the test range.
2. Build the expected table with =$D5*B$8/D8 (row total × column total / grand total).
3. Compute the contributions =(B5−B11)^2/B11 and add them to get $\chi^2 = 9.002$.
4. =CHISQ.TEST(B5:C7,B11:C13) returns the p-value 0.0111. Compare with $\alpha = 0.05$.

3.5 Python code

```
import numpy as np
from scipy.stats import chi2_contingency

# Rows: Greetings, Medical, Numbers; columns: Correct, Incorrect
observed = np.array([[93, 7],
                     [85, 15],
                     [78, 22]])

chi2, p_value, df, expected = chi2_contingency(observed, correction=False)
print("Expected counts:\n", expected)
print("Chi-square:", chi2, "df:", df, "p-value:", p_value)

cramers_v = np.sqrt(chi2 / (observed.sum() * (min(observed.shape) - 1)))
print("Cramer V:", cramers_v)

if p_value < 0.05:
    print("Reject H0: evidence of an association.")
else:
    print("Fail to reject H0: insufficient evidence of an association.")
```

3.6 Result and interpretation (on the assumed data)

$\chi^2(2, N = 300) = 9.00$, $p = .011$, Cramer's $V = 0.17$. All expected counts (85.33 and 14.67) are above 5, so the chi-square approximation is valid.

Since $p < 0.05$, **reject H0**: recognition outcome is associated with sign category. The error rate rises from 7% for Greetings to 15% for Medical and 22% for Numbers. The standardised residuals show that Greetings have fewer errors than expected (−2.00) and Numbers have more than expected (+1.91, just under the 1.96 cut-off). The effect size is small to moderate ($V = 0.17$), so the category explains only a limited part of the errors.

Suggested reporting sentence: "A chi-square test of independence showed a significant association between sign category and recognition outcome, $\chi^2(2, N = 300) = 9.00$, $p = .011$, $V = .17$, with error rates of 7% (Greetings), 15% (Medical) and 22% (Numbers)."

3.7 Assumptions and cautions

- Both variables are categorical and each attempt falls in exactly one cell.
- **Independence:** attempts are assumed independent. In real testing the same signers repeat many attempts, so observations are clustered and a mixed-effects model would be safer.
- The test shows association, not the cause. Numbers may be harder because many digits share similar hand shapes, but this needs separate evidence.

4. Quick Comparison of the Tests Used

Feature	Welch t-test	One-way ANOVA	Chi-square
Question here	Baseline vs proposed: false outputs per minute	Accuracy across 3 lighting conditions	Sign category vs correct / incorrect
Variable types	Numerical by 2 groups	Numerical by 3 groups	Two categorical
Statistic	$t(15.75) = 5.81$	$F(2, 21) = 24.46$	$\chi^2(2) = 9.00$
p-value	< .001 (0.000029)	< .001 (0.0000033)	0.011
Effect size	$d = 2.60$	$\eta^2 = 0.70$	$V = 0.17$
Decision ($\alpha = 0.05$)	Reject H0	Reject H0 (Tukey: all pairs differ)	Reject H0
Excel tool	t-Test: Unequal Variances	ANOVA: Single Factor	CHISQ.TEST
Python	stats.ttest_ind(equal_var=False)	stats.f_oneway	chi2_contingency

5. Overall Interpretation for the Project

On the assumed data, all three tests reject their null hypotheses, and each links to a gap identified in our SLR. (1) The Idle class with debounce filter roughly halves false outputs during idle periods (gap G3). (2) Accuracy depends strongly on lighting, falling almost 7 points from Bright to Dim (gap G2, domain shift). (3) Some sign categories, such as Numbers, are recognised less reliably than others, which points to where more training data should be collected.

These are statements about the **assumed** samples. They show how the tests would be used on BhashaBridge, and what pattern of real results would support the design decisions. They are not evidence that the real system already performs this way.

6. Limitations

- All data are assumed for the exercise; no real recognition experiment has been run yet.
- Sample sizes are small (10, 8 and 100 per group), and no power analysis was done.
- Independence of observations is assumed. Real sessions from the same signers would need repeated-measures or mixed-effects methods.
- Each test is applied to a separate dataset, so no multiple-testing correction was needed. If several comparisons were made on one dataset, a Bonferroni correction would be applied.
- Statistical significance is not practical significance; accuracy and latency must also meet real-time requirements for deaf users.

7. Future Work to Replace the Assumed Data

- Record idle sessions, lighting-varied sessions and sign-category attempts as planned in the SLR validation plan, with informed consent as set out in Assignment 7 (Ethics).
- Use signer-independent splits so that results are not inflated by repeated signers.
- Replace the arrays in the attached Python script or the Excel workbook with the measured values; no other change is needed.
- For paired model comparisons on the same test clips, use McNemar’s test (two models) or Cochran’s Q (three models).
- Run a power analysis to decide how many sessions are needed before the real experiment.